

Expository Notes on Dynamical Systems

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1 Introduction

The purpose of these notes is to explore several ideas that naturally arise from the perspective of dynamical systems. Beginning with simple one-dimensional system and classical projectile motion, the discussion gradually develops concepts such as phase portraits, conservation laws, linearization, and chaos. The emphasis is placed on intuition, geometric interpretation, and the connections between seemingly many different ideas.

Most of the results presented here are standard, while few others are motivated by personal observations and numerical experiments. Throughout these notes, the goal is not simply to derive mathematical results, but to understand why they arise and what they reveal about the underlying dynamical systems.

2 When a One-Dimensional System Isn't Quite One-Dimensional

2.1 Model and Derivation

Consider a point mass m moving radially under gravity from a fixed mass M (Earth). Using Newton's second law, we can write;

$$m\ddot{r} = -\frac{GMm}{r^2}.$$

Multiply both sides by $\dot{r}$:

$$m\ddot{r}\dot{r} = -\frac{GMm}{r^2}\dot{r}.$$

Now integrate both sides with respect to time:

$$\int m\ddot{r}\dot{r} dt = \int -\frac{GMm}{r^2}\dot{r} dt.$$

On the left-hand side, observe that $\ddot{r} dt = d\dot{r}$, so

$$\int m\ddot{r}\dot{r} dt = \int m\dot{r} d\dot{r}.$$

On the right-hand side, note that $\dot{r} dt = dr$, so

$$\int -\frac{GMm}{r^2}\dot{r} dt = \int -\frac{GMm}{r^2} dr.$$

Evaluating both integrals gives

$$\frac{1}{2}m\dot{r}^2 - \frac{GMm}{r} = C,$$

where C is a constant of integration, which matches with our intuition from physics. Interpreting the terms, the first term is kinetic energy, the second is gravitational potential energy, so C is the conserved total mechanical energy. This proves the Law of conservation of Mechanical Energy.

Now, replacing c with E , we have;

$$E = \frac{1}{2}m\dot{r}^2 - \frac{GMm}{r}.$$

where E is the constant total mechanical energy set by the launch.

2.2 Energy Changes Everything

Up to this point, the problem is simply a second-order differential equation. Once mechanical energy is introduced, however, the focus shifts completely. Rather than asking for an explicit solution, the conserved energy allows us to ask qualitative questions about the motion itself. Which regions of space are physically accessible? Does the particle eventually return, or does it escape the gravitational field altogether? Where can the motion reverse direction? The conserved energy already contains much of this information before the differential equation is ever solved.

Solving for $\dot{r}$ gives not just one, but two first-order autonomous differential equations corresponding to the positive and negative branches,

$$\dot{r} = f(r) = \pm \sqrt{\frac{2}{m} \left(E + \frac{GMm}{r} \right)}, \quad r > 0, \quad E + \frac{GMm}{r} \geq 0.$$

At first glance, this appears to be an ordinary one-dimensional autonomous differential equation. However, it is not. The square root naturally introduces two possible vector fields. The positive branch describes outward motion, while the negative branch describes inward motion. This creates an interesting conceptual situation. The model is not genuinely two-dimensional, since the only state variable is still r . Yet it is also not an ordinary one-dimensional autonomous flow, since a single value of r no longer determines a unique direction of motion.

2.3 Physical Domain and Admissible Motion

The energy equation is

$$E = \frac{1}{2}m\dot{r}^2 - \frac{GMm}{r}.$$

The gravitational potential energy satisfies

$$-\frac{GMm}{r} \rightarrow 0 \quad \text{as} \quad r \rightarrow \infty.$$

Therefore, any trajectory that reaches arbitrarily large radii must satisfy

$$E = \lim_{r \rightarrow \infty} \frac{1}{2}m\dot{r}^2 \geq 0.$$

This single observation separates the motion into three distinct energy regimes.

2.3.1 Positive Energy

Suppose $E > 0$. Then

$$E + \frac{GMm}{r} > 0$$

for every $r > 0$, since both terms are positive. Consequently, the square root remains real for every positive radius, so the energy equation places no upper bound on the motion.

To understand what this means physically, recall that the total mechanical energy is

$$E = \frac{1}{2}m\dot{r}^2 - \frac{GMm}{r}.$$

As the particle moves farther away from the central mass, the gravitational potential energy

$$-\frac{GMm}{r}$$

becomes less negative and approaches zero. Since the total mechanical energy is conserved and remains strictly positive, the kinetic energy must approach the same positive value,

$$\frac{1}{2}m\dot{r}^2 \longrightarrow E > 0.$$

In other words, the particle never exhausts its kinetic energy. Gravity continuously slows the particle as it moves outward, but it is never able to reduce the speed to zero. Consequently, no turning point can occur, and the particle continues moving away from the central mass indefinitely.

2.3.2 Negative Energy

Suppose $E < 0$. From the observation above, a trajectory with negative total energy cannot reach infinity, because doing so would require its limiting kinetic energy to equal the negative number E .

The admissibility condition

$$E + \frac{GMm}{r} \geq 0$$

gives

$$r \leq -\frac{GMm}{E} = \frac{GMm}{|E|}.$$

Hence the motion is restricted to

$$0 < r \leq r_{\max}, \quad r_{\max} = \frac{GMm}{|E|}.$$

At $r = r_{\max}$,

$$\dot{r} = 0.$$

The particle is momentarily at rest, but

$$\ddot{r} = -\frac{GM}{r_{\max}^2} < 0,$$

so it accelerates inward, reverses direction, and falls back toward the central mass.

2.3.3 Zero Energy

Finally, suppose $E = 0$. Then

$$\frac{1}{2}m\dot{r}^2 = \frac{GMm}{r},$$

so

$$\dot{r}^2 = \frac{2GM}{r}.$$

For every finite $r > 0$, the right-hand side is positive. Therefore the radial speed never vanishes at a finite radius. However,

$$\dot{r}^2 \longrightarrow 0 \quad \text{as} \quad r \longrightarrow \infty.$$

Thus the particle continues moving outward for arbitrarily large distances while slowing indefinitely. It reaches zero speed only in the limit at infinity. This is the threshold case separating bounded motion from escape with positive limiting speed.

2.4 Existence, Uniqueness, and an Interesting Failure of Uniqueness

The vector field

$$f(r) = \pm \sqrt{\frac{2}{m} \left(E + \frac{GMm}{r} \right)}$$

is continuous throughout the admissible domain described in the previous section. Therefore, by the Existence Theorem, a solution exists through every admissible initial condition.

For the function

$$f(r) = \pm \sqrt{\frac{2}{m} \left(E + \frac{GMm}{r} \right)},$$

The derivative is,

$$f'(r) = \pm \frac{GM}{r^2 \sqrt{\frac{2}{m} \left(E + \frac{GMm}{r} \right)}}.$$

Whenever

$$E + \frac{GMm}{r} > 0,$$

the denominator remains finite. Consequently, f is locally Lipschitz on every closed interval contained in the admissible region. By the Existence and Uniqueness Theorem, the solution is locally unique. Physically, this means that once the initial position and direction of motion are specified, the future trajectory is completely determined.

Something interesting happens at the turning point. Recall from the previous section that for $E < 0$ the particle reaches the turning point

$$r_{\max} = \frac{GMm}{|E|},$$

where

$$\dot{r} = 0.$$

At this point,

$$E + \frac{GMm}{r} = 0,$$

so the denominator of

$$f'(r)$$

vanishes.

The vector field remains continuous, so existence is still guaranteed. However, it is no longer locally Lipschitz, and the standard uniqueness theorem no longer applies.

What is fascinating is that this mathematical failure occurs precisely at the most physically meaningful point of the motion. The turning point is the instant at which the outward trajectory comes momentarily to rest before reversing direction.

From the perspective of mathematical analysis, this is where the hypotheses of the uniqueness theorem fail. From the perspective of physics, it is simply the highest point of the trajectory. The mathematics and the physics are describing the same phenomenon from two different viewpoints.

3 State Space Is Not Physical Space

In previous chapter, analysis focused entirely on the radial motion, which allowed us to understand how the distance from the central mass changes in time, including turning points and the distinction between inward and outward motion. Physically, this matched the intuition that a particle may move away from the Earth or fall toward it, and the “ $\pm$ ” branches captured these two behaviors. However, as noted in the first chapter, the system was not truly one-dimensional. Analyzing $r(t)$ only tells us how far the particle is from the origin but not where it is in the plane. Like two different objects could be equidistant from the origin, but actually at completely different locations in space. Without tracking the angular variable, the description of the motion is incomplete. So, we are now incorporating the angular component of the motion as well.

3.1 Deriving the Planar Model

To describe the motion of the particle in the plane, begin with in polar coordinates using the functions

$$r(t), \quad \theta(t),$$

which together determine the position of the particle. Acceleration in polar form can be obtained by differentiating the velocity components.

$$a_r = \ddot{r} - r\dot{\theta}^2, \quad a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}.$$

The term a_r measures the change in the radial speed after accounting for angular motion, and a_θ measures the angular acceleration corrected by the radial change.

Newton’s laws in polar components. Newton’s second law states that m times the acceleration equals the force. Because gravitational attraction acts purely in the radial direction with magnitude GMm/r^2 , the angular component of the force is zero. Thus

$$ma_r = -\frac{GMm}{r^2}, \quad ma_\theta = 0.$$

Substituting the polar acceleration formulas into these two equations gives

$$m(\ddot{r} - r\dot{\theta}^2) = -\frac{GMm}{r^2},$$

$$m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = 0.$$

Dividing by m and rearranging:

$$r\ddot{\theta} + 2\dot{r}\dot{\theta} = 0.$$

Multiplying by r converts it into a perfect derivative:

$$r^2\ddot{\theta} + 2r\dot{r}\dot{\theta} = \frac{d}{dt}(r^2\dot{\theta}) = 0,$$

Now, integrating on both sides, we get

$$r^2\dot{\theta} = \ell,$$

where ℓ is the constant angular momentum of the motion. Solving for $\dot{\theta}$ yields the angular differential equation

$$\dot{\theta} = \frac{\ell}{r^2}.$$

Substituting this expression into the radial equation derived earlier, gives

$$\ddot{r} - r \left(\frac{\ell^2}{r^4} \right) = -\frac{GM}{r^2},$$

which simplifies to

$$\ddot{r} = \frac{\ell^2}{r^3} - \frac{GM}{r^2}.$$

So, We have these following equations:

$$\boxed{\begin{aligned}\dot{r} &= v, \\ \dot{v} &= \frac{\ell^2}{r^3} - \frac{GM}{r^2}, \\ \dot{\theta} &= \frac{\ell}{r^2}\end{aligned}}$$

3.2 State Space is not physical space

At first glance, the system appears to have become three-dimensional. Has the particle somehow left the plane? Of course not.

Although the polar-coordinate formulation produces three first-order equations, this does not necessarily mean the particle is moving in three physical dimensions. The state-space dimension of a differential equation is not the same as the physical dimension of the motion. Physically the orbit lies in a two-dimensional plane, but when the second-order radial equation is rewritten as a first-order system by introducing $v = \dot{r}$, the number of dynamical variables increases. This is a mathematical thing rather than a physical statement.

The angular variable θ plays a different role. Its evolution depends on the radial motion through $r(t)$, but it does not influence the equations governing r and v . In other words, $\dot{\theta}$ is a like bookkeeping equation that records how the orbit sweeps around the origin, while the actual nonlinear dynamics occur entirely in the (r, v) plane. Because of this reason, the $\dot{\theta}$ equation is omitted in the phase-portrait analysis. The angular equation will reappear only when establishing Kepler's second law in the following subsection.

3.3 Kepler's Laws from the Angular Equation

The area swept out by the radius vector during a small angular change $d\theta$ is the area of a circular sector:

$$dA = \frac{1}{2}r^2 d\theta.$$

The instantaneous area sweep rate

$$\frac{dA}{dt} = \frac{1}{2} r^2 \dot{\theta}.$$

Using the angular momentum conservation law $r^2 \dot{\theta} = \ell$, this becomes

$$\frac{dA}{dt} = \frac{\ell}{2},$$

which is constant. Thus the orbit sweeps out equal areas in equal times, establishing Kepler's Second Law directly from the polar-coordinate formulation. Kepler's other laws are beyond the scope of this note.

3.4 Nullclines and Equilibrium of the reduced planar system

The r -nullcline is obtained by setting $\dot{r} = 0$:

$$\dot{r} = v = 0 \implies v = 0.$$

Thus the r -nullcline is the horizontal axis $v = 0$ in the (r, v) -plane.

The v -nullcline is obtained by setting $\dot{v} = 0$:

$$\dot{v} = \frac{\ell^2}{r^3} - \frac{GM}{r^2} = 0.$$

Assuming $r > 0$, this simplifies to

$$\frac{\ell^2}{r^3} = \frac{GM}{r^2} \implies r = \frac{\ell^2}{GM} = r_0(\text{let}).$$

Hence the v -nullcline is the vertical line $r = r_0$. The unique equilibrium point in the physical domain $r > 0$ is the intersection of the two nullclines:

$$(r_0, v_0) = \left(\frac{\ell^2}{GM}, 0 \right).$$

3.5 Linearization Near the Equilibrium

To study the local behavior near this point, the Jacobian matrix of the vector field is evaluated at (r_0, v_0) .

Writing

$$f(r, v) = v, \quad g(r, v) = \frac{\ell^2}{r^3} - \frac{GM}{r^2},$$

the Jacobian is

$$J(r, v) = \begin{pmatrix} f_r & f_v \\ g_r & g_v \end{pmatrix}.$$

From $f(r, v) = v$ it follows that

$$f_r = 0, \quad f_v = 1.$$

From

$$g(r, v) = \frac{\ell^2}{r^3} - \frac{GM}{r^2}$$

Differentiating with respect to r and v at equilibrium point gives

$$g_r(r) = -3\ell^2 r^{-4} + 2GM r^{-3} = -\frac{(GM)^4}{\ell^6},$$

and

$$g_v = 0,$$

since g does not depend on v .

Thus the Jacobian at the equilibrium is

$$J(r_0, v_0) = \begin{pmatrix} 0 & 1 \\ -\frac{(GM)^4}{\ell^6} & 0 \end{pmatrix}.$$

The characteristic polynomial is

$$\lambda^2 + \frac{(GM)^4}{\ell^6} = 0,$$

so the eigenvalues are purely imaginary:

$$\lambda = \pm i \frac{(GM)^2}{\ell^3}.$$

Therefore

$$\text{tr } J = 0, \quad \det J = \frac{(GM)^4}{\ell^6} > 0,$$

From the analysis of eigenvalues, the equilibrium $(r_0, 0)$ is classified as a center. A center is stable (nearby trajectories remain near the equilibrium), but it is (not asymptotically stable), since the eigenvalues have zero real part. Trajectories do not spiral inward, instead they form closed periodic orbits.

3.6 Phase Portrait and Center Dynamics

Now that we know equilibrium is a center. This classification is much easier to appreciate through the phase portrait. As shown in Figure 1, the vector field exhibits a clear rotational pattern around the equilibrium. Nearby trajectories circulate around the equilibrium without spiraling inward or outward, forming a family of closed curves. This behavior is consistent with the classification obtained from the eigenvalue analysis: the equilibrium is stable in the Lyapunov sense, but not asymptotically stable.

Although the phase portrait strongly suggests periodic motion, it does not explain why the trajectories remain on closed curves. That question will become important in the next section, where the limitations of linearization are discussed.

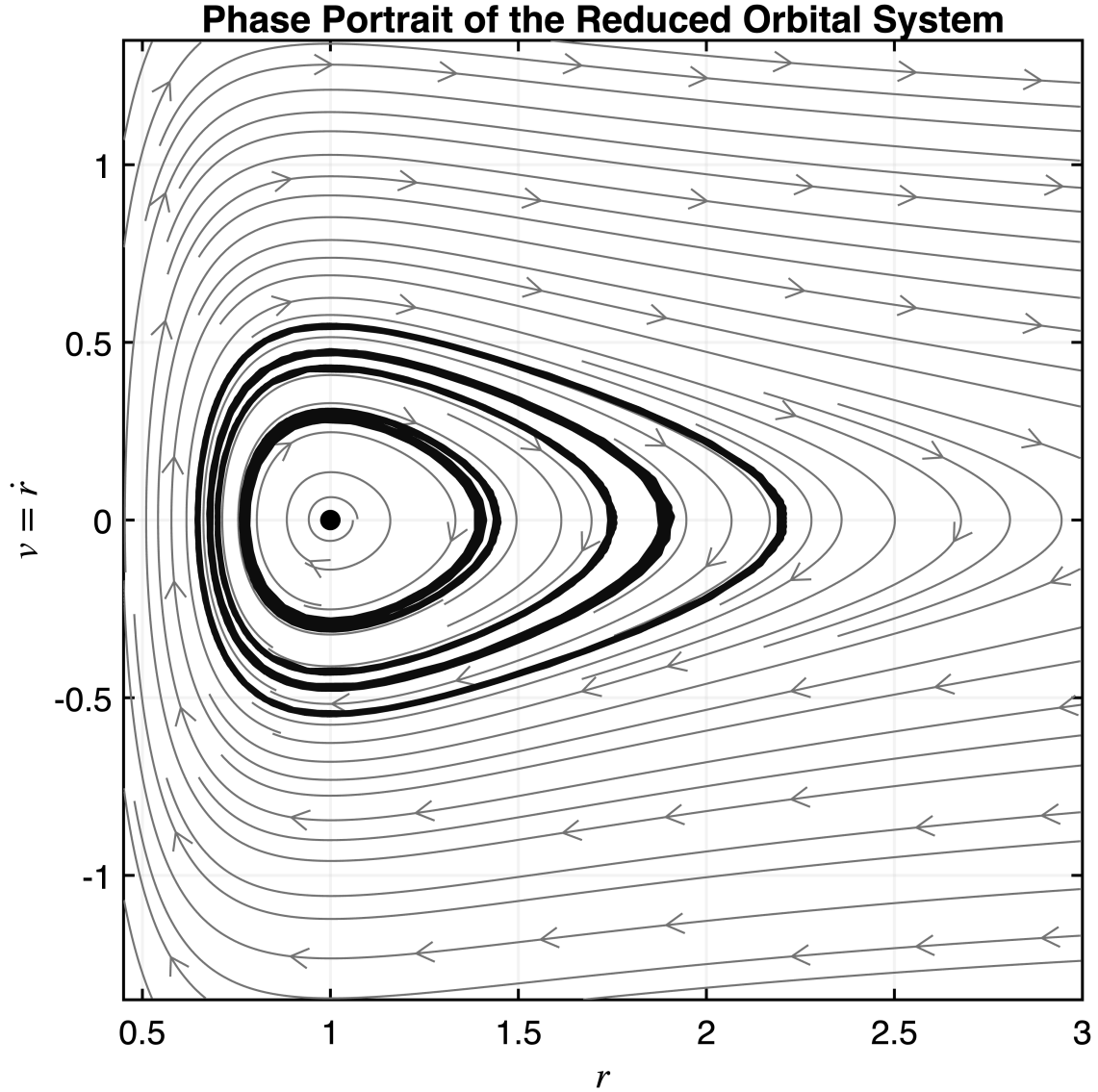


Figure 1: Phase Potrait in the r-v plane

3.7 Hartman–Grobman and Its Failure

This is precisely where one of the most fundamental theorems in dynamical systems becomes relevant. The **Hartman–Grobman Theorem** states that, in a neighborhood of a hyperbolic equilibrium point, a nonlinear system is topologically equivalent to its linearization. In other words, the local behavior of the nonlinear system can be understood by studying the Jacobian matrix of the linearized system.

The key word is *hyperbolicity*. An equilibrium is hyperbolic only if none of the eigenvalues of the Jacobian have zero real part.

For the present system, the eigenvalues are

$$\lambda = \pm i \frac{(GM)^2}{\ell^3}.$$

which are purely imaginary. Their real parts are therefore exactly zero. Consequently, the

equilibrium is *non-hyperbolic*, and the Hartman–Grobman Theorem does not apply.

This means that the phase portrait obtained above cannot be justified solely from the linearization. The Jacobian correctly predicts rotational behavior near the equilibrium, but it does not explain why nearby trajectories remain on closed curves rather than spiraling inward or outward. To understand that structure, an additional idea is needed.

That idea comes from *conservation*. The system possesses a conserved energy, and as we shall see in the next section, the geometry of the phase portrait is ultimately determined by the level sets of that energy function rather than by the linearization alone.

3.8 Conservation Creates Geometry

Since the Hartman–Grobman Theorem does not apply, linearization alone cannot explain the closed curves in the phase portrait. To understand why the trajectories remain trapped on loops, we need another source of structure. A natural question is whether the system possesses a conserved quantity.

Starting from the reduced system,

$$\begin{aligned}\dot{r} &= v, \\ \dot{v} &= \frac{\ell^2}{r^3} - \frac{GM}{r^2},\end{aligned}$$

we eliminate time by writing

$$\frac{dv}{dr} = \frac{\dot{v}}{\dot{r}},$$

which gives

$$\frac{dv}{dr} = \frac{\frac{\ell^2}{r^3} - \frac{GM}{r^2}}{v}.$$

Multiplying through by $v \, dr$ yields

$$v \, dv = \left(\frac{\ell^2}{r^3} - \frac{GM}{r^2} \right) dr.$$

Rearranging,

$$- \left(\frac{\ell^2}{r^3} - \frac{GM}{r^2} \right) dr + v \, dv = 0.$$

This has the form

$$M(r, v) \, dr + N(r, v) \, dv = 0,$$

where

$$M(r, v) = -\frac{\ell^2}{r^3} + \frac{GM}{r^2},$$

and

$$N(r, v) = v.$$

The exactness condition requires

$$\frac{\partial M}{\partial v} = \frac{\partial N}{\partial r}.$$

Here,

$$\frac{\partial M}{\partial v} = 0, \quad \frac{\partial N}{\partial r} = 0,$$

so the differential form is exact. Therefore there exists a scalar function

$$E(r, v)$$

such that

$$dE = M dr + N dv.$$

To determine E , integrate M with respect to r :

$$E(r, v) = \int \left(-\frac{\ell^2}{r^3} + \frac{GM}{r^2} \right) dr.$$

Carrying out the integration gives

$$E(r, v) = \frac{\ell^2}{2r^2} - \frac{GM}{r} + g(v).$$

Differentiating with respect to v ,

$$\frac{\partial E}{\partial v} = g'(v).$$

Since

$$\frac{\partial E}{\partial v} = N(r, v) = v,$$

we obtain

$$g'(v) = v.$$

Integrating once more,

$$g(v) = \frac{1}{2}v^2.$$

Hence the conserved quantity is

$$E(r, v) = \frac{1}{2}v^2 + \frac{\ell^2}{2r^2} - \frac{GM}{r}.$$

The significance of this calculation is not merely that a conserved quantity exists. It explains the geometry of the phase portrait.

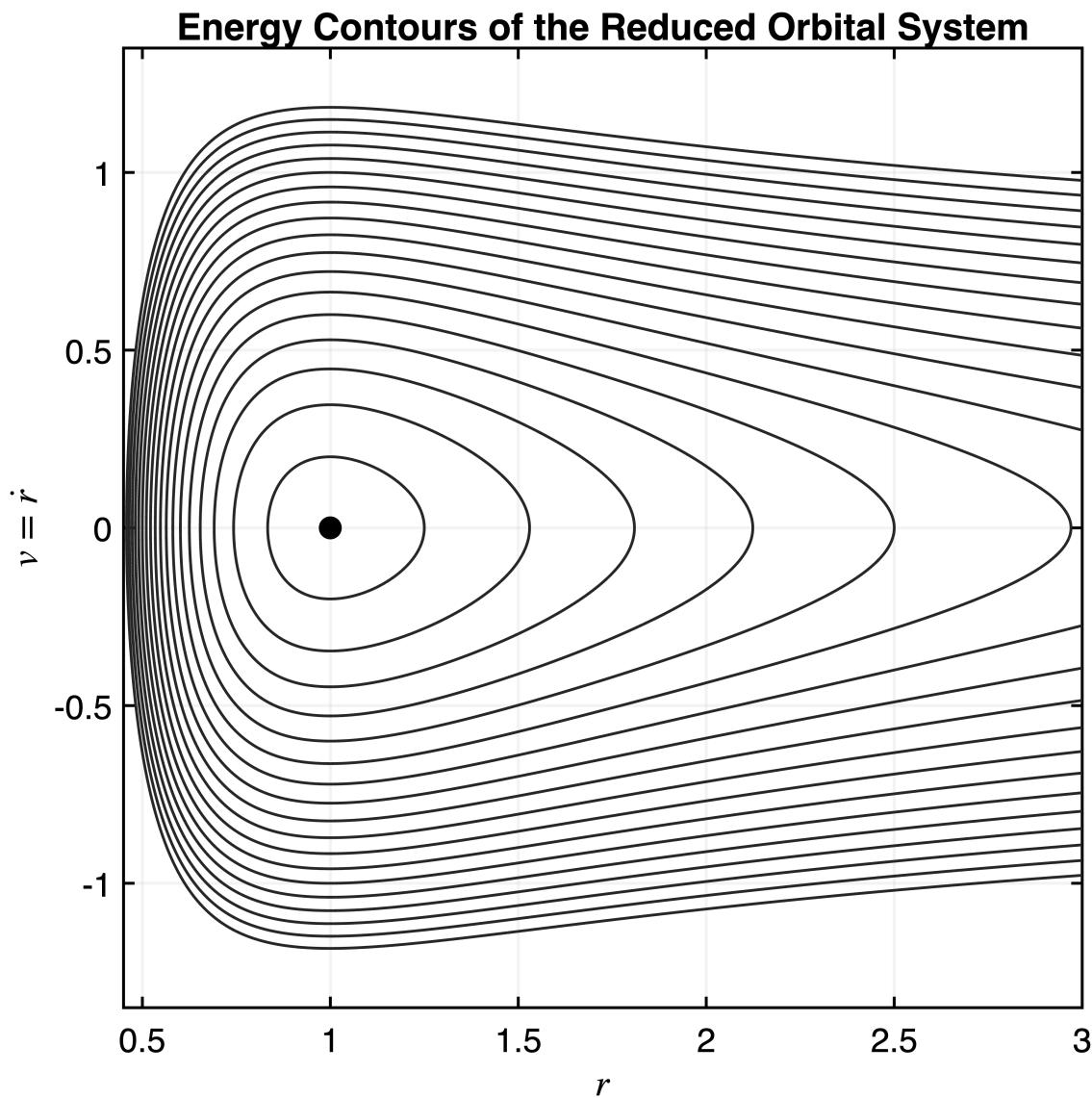


Figure 2: Energy Contours of reduced orbital system

Since $E(r, v)$ remains constant along trajectories, solutions are confined to the level sets of E . The closed curves observed in the phase portrait are therefore not accidental. They are a direct consequence of the conserved structure of the system.

This also makes the earlier phase portrait much less mysterious. Because $E(r, v)$ is constant along every solution, trajectories cannot move freely throughout the (r, v) -plane. Instead, they are constrained to remain on the level curves as shown in Figure 2 above.

3.9 Limit Cycles and the Poincaré–Bendixson Theorem

Figure 2 also explains why the closed curves in the phase portrait are *not* limit cycles. A limit cycle is an isolated periodic orbit that attracts or repels nearby trajectories. Here, this cannot occur because every solution remains confined to its own energy level set. Since the energy is conserved, trajectories cannot gradually spiral toward a preferred orbit or drift away from it. Instead, each initial condition determines a different closed curve, and the motion remains on that curve for all

time.

The distinction is subtle but important. The closed trajectories surrounding the equilibrium are indeed periodic orbits, but they are not limit cycles. Nothing attracts nearby trajectories, and nothing repels them. The family of closed curves arises entirely from conservation of energy.

The planar nature of the system imposes an additional restriction. The **Poincaré–Bendixson Theorem** states that bounded trajectories in two-dimensional systems can only approach a limited class of long-term behaviors, namely equilibria or periodic orbits.

In the present setting, however, the theorem is almost "secondary". The conservation law has already determined the geometry of the motion. Since every trajectory is trapped on an energy level set, the phase portrait already consists of closed periodic orbits. The Poincaré–Bendixson Theorem simply confirms that no more complicated long-term behavior can arise.

3.10 Lyapunov Stability of the Equilibrium

The conserved energy function provides a natural Lyapunov function for the equilibrium. Since the equilibrium does not occur at zero energy, it is convenient to shift the energy by a constant and define

$$V(r, v) = E(r, v) - E(\hat{r}, 0).$$

This shift changes neither the physics nor the geometry of the system. It simply normalizes the equilibrium so that

$$V(\hat{r}, 0) = 0.$$

Now, to justify the positivity of V , we show that the equilibrium is a strict local minimum of the energy.

The partial derivatives of the energy are

$$\frac{\partial E}{\partial r} = -\frac{\ell^2}{r^3} + \frac{GM}{r^2},$$

and

$$\frac{\partial E}{\partial v} = v.$$

At the equilibrium,

$$(\hat{r}, 0),$$

we have

$$\frac{\partial E}{\partial r}(\hat{r}, 0) = 0, \quad \frac{\partial E}{\partial v}(\hat{r}, 0) = 0,$$

so the equilibrium is a critical point of the energy.

To classify this critical point, compute the Hessian matrix:

$$D^2E(r, v) = \begin{pmatrix} \frac{3\ell^2}{r^4} - \frac{2GM}{r^3} & 0 \\ 0 & 1 \end{pmatrix}.$$

Using the equilibrium relation

$$\hat{r} = \frac{\ell^2}{GM},$$

or equivalently,

$$\ell^2 = GM\hat{r},$$

gives

$$\begin{aligned} \frac{3\ell^2}{\hat{r}^4} - \frac{2GM}{\hat{r}^3} &= \frac{3GM\hat{r}}{\hat{r}^4} - \frac{2GM}{\hat{r}^3} \\ &= \frac{GM}{\hat{r}^3}. \end{aligned}$$

Hence,

$$D^2E(\hat{r}, 0) = \begin{pmatrix} \frac{GM}{\hat{r}^3} & 0 \\ 0 & 1 \end{pmatrix}.$$

Since both diagonal entries are strictly positive, the Hessian is positive definite. Therefore, $(\hat{r}, 0)$ is a strict local minimum of the energy.

Consequently,

$$V(r, v) = E(r, v) - E(\hat{r}, 0) \geq 0,$$

with equality if and only if

$$(r, v) = (\hat{r}, 0).$$

Thus, V is positive definite in a neighborhood of the equilibrium.

Thus, equilibrium corresponds to the minimum value of the energy. Consequently,

$$V(r, v) \geq 0$$

for all points sufficiently close to the equilibrium, with equality if and only if

$$(r, v) = (\hat{r}, 0).$$

Thus, V is positive definite in a neighborhood of the equilibrium. Since the total energy is conserved,

$$\dot{V} = 0$$

along every solution. The value of V therefore remains constant as the system evolves.

Consequently, the equilibrium is *Lyapunov stable*: solutions that begin sufficiently close to the equilibrium remain close for all future time.

The equilibrium is *not* asymptotically stable, however. Because the energy remains constant, trajectories cannot lose energy and move toward the equilibrium as time increases. Instead, they remain confined to the surrounding energy contours and continue circulating indefinitely. This is precisely the behavior expected of a nonlinear center.

At this point, the qualitative picture of the planar system is essentially complete. Energy conservation explains the geometry of the phase portrait, Lyapunov theory explains the stability of the equilibrium, and the Poincaré–Bendixson Theorem explains why no more complicated long-term behavior can occur.

4 Where Does Chaos Come From?

The previous chapter showed that planar dynamical systems are strongly constrained. This naturally raises a new question: if two-dimensional systems are so structured and predictable, where does chaos come from?

Something fundamentally different must happen when the state space has three or more dimensions. The additional dimension allows trajectories to interact in ways that are impossible in a planar system, producing much richer long-term behavior.

4.1 The Lorenz System

One of the most famous examples of chaotic dynamics is the Lorenz system,

$$\begin{aligned}\dot{x} &= \sigma(y - x), \\ \dot{y} &= x(\rho - z) - y, \\ \dot{z} &= xy - \beta z.\end{aligned}$$

The system was originally introduced by Edward Lorenz as a simplified model of atmospheric convection. Despite consisting of only three nonlinear differential equations, it can produce remarkably complicated long-term behavior.

Unlike the planar systems studied in the previous chapter, the Lorenz system evolves in a three-dimensional state space. Many of the restrictions that apply to two-dimensional systems no longer hold, allowing more complicated dynamics to appear.

One feature of the system is especially striking. The equations are completely deterministic, yet their long-term behavior can be extremely difficult to predict.

4.2 Sensitive Dependence on Initial Conditions

One of the main features associated with chaotic systems is *sensitive dependence on initial conditions*. Two trajectories that begin extremely close together may eventually become widely separated, even though both satisfy exactly the same differential equations.

Consider two nearby solutions,

$$X_1(t) \quad \text{and} \quad X_2(t).$$

Their separation is represented by

$$\delta X(t) = X_2(t) - X_1(t).$$

In the Lorenz system, however, this small difference may grow rapidly over time. This is the mathematical idea behind the butterfly effect.

At first, this appears to provide a complete explanation of chaotic unpredictability: small differences grow, so long-term prediction becomes unreliable. However, this description led to another question. What would happen if the growth of the separation were deliberately opposed?

The following numerical experiment was designed to explore that question.

4.3 Tracking the Perturbation

Two nearby Lorenz trajectories were considered. Their instantaneous difference was calculated at every numerical time step:

$$\delta X(t) = X_2(t) - X_1(t).$$

The first trajectory was allowed to evolve according to the original Lorenz equations. A correction based on the separation was then subtracted from the second trajectory.

Thus, the two trajectories satisfy

$$\dot{X}_1 = f(X_1),$$

and

$$\dot{X}_2 = f(X_2) - C,$$

where C is a correction constructed from the current difference between X_1 and X_2 .

The purpose of the correction was simply to oppose the separation of the two trajectories and observe how the resulting motion changed.

4.4 Recursive Feedback

Instead of applying the perturbation correction only once during each time step, the correction was built through several recursive stages.

Let

$$\delta X_0 = \delta X, \quad C_0 = 0,$$

where δX is the instantaneous difference between the two trajectories,

$$\delta X = X_2 - X_1.$$

The recursive updates are

$$C_{j+1} = C_j + \alpha \delta X_j,$$

and

$$\delta X_{j+1} = \delta X_j - \alpha \Delta t \delta X_j,$$

$$\delta X_{j+1} = (1 - \alpha \Delta t) \delta X_j.$$

for

$$j = 0, 1, \dots, K - 1.$$

The first recursive stage gives

$$\delta X_1 = (1 - \alpha \Delta t) \delta X_0,$$

and

$$\begin{aligned} C_1 &= C_0 + \alpha \delta X_0 \\ &= 0 + \alpha \delta X_0 \\ &= \alpha \delta X_0. \end{aligned}$$

At the second recursive stage,

$$\begin{aligned}\delta X_2 &= (1 - \alpha\Delta t)\delta X_1 \\ &= (1 - \alpha\Delta t) [(1 - \alpha\Delta t)\delta X_0] \\ &= (1 - \alpha\Delta t)^2\delta X_0.\end{aligned}$$

The correction becomes

$$\begin{aligned}C_2 &= C_1 + \alpha\delta X_1 \\ &= \alpha\delta X_0 + \alpha(1 - \alpha\Delta t)\delta X_0 \\ &= \alpha [1 + (1 - \alpha\Delta t)] \delta X_0.\end{aligned}$$

At the third recursive stage,

$$\begin{aligned}\delta X_3 &= (1 - \alpha\Delta t)\delta X_2 \\ &= (1 - \alpha\Delta t) [(1 - \alpha\Delta t)^2\delta X_0] \\ &= (1 - \alpha\Delta t)^3\delta X_0.\end{aligned}$$

The correction becomes

$$\begin{aligned}C_3 &= C_2 + \alpha\delta X_2 \\ &= \alpha [1 + (1 - \alpha\Delta t)] \delta X_0 + \alpha(1 - \alpha\Delta t)^2\delta X_0 \\ &= \alpha [1 + (1 - \alpha\Delta t) + (1 - \alpha\Delta t)^2] \delta X_0.\end{aligned}$$

After j recursive stages,

$$\boxed{\delta X_j = (1 - \alpha\Delta t)^j\delta X_0.}$$

Similarly, after K stages, the accumulated correction is

$$C_K = \alpha\delta X_0 + \alpha\delta X_1 + \alpha\delta X_2 + \cdots + \alpha\delta X_{K-1}.$$

Substituting

$$\delta X_j = (1 - \alpha\Delta t)^j\delta X_0$$

into each term gives

$$\begin{aligned}C_K &= \alpha\delta X_0 + \alpha(1 - \alpha\Delta t)\delta X_0 \\ &\quad + \alpha(1 - \alpha\Delta t)^2\delta X_0 + \cdots \\ &\quad + \alpha(1 - \alpha\Delta t)^{K-1}\delta X_0.\end{aligned}$$

Factoring out $\alpha\delta X_0$,

$$C_K = \alpha [1 + (1 - \alpha\Delta t) + (1 - \alpha\Delta t)^2 + \cdots + (1 - \alpha\Delta t)^{K-1}] \delta X_0.$$

For a finite geometric series,

$$1 + r + r^2 + \cdots + r^{K-1} = \frac{1 - r^K}{1 - r}.$$

So, We get:

$$\begin{aligned}
1 + (1 - \alpha\Delta t) + \cdots + (1 - \alpha\Delta t)^{K-1} &= \frac{1 - (1 - \alpha\Delta t)^K}{1 - (1 - \alpha\Delta t)} \\
&= \frac{1 - (1 - \alpha\Delta t)^K}{\alpha\Delta t}.
\end{aligned}$$

Substituting this expression into the formula for C_K gives

$$C_K = \alpha \left[\frac{1 - (1 - \alpha\Delta t)^K}{\alpha\Delta t} \right] \delta X_0.$$

Hence,

$$C_K = \frac{1 - (1 - \alpha\Delta t)^K}{\Delta t} \delta X_0.$$

Define

$$\gamma_K = \frac{1 - (1 - \alpha\Delta t)^K}{\Delta t}.$$

Then the correction may be written compactly as

$$C_K = \gamma_K \delta X.$$

5 Numerical Experiments and Simulations

The recursive perturbation feedback developed in the previous section was implemented in MATLAB using the explicit Euler method. Throughout the simulations, the classical Lorenz parameters were fixed as

$$\sigma = 10, \quad \rho = 28, \quad \beta = \frac{8}{3},$$

with time step

$$\Delta t = 0.001,$$

and simulation interval

$$0 \leq t < 200.$$

Two nearby initial conditions were chosen,

$$X_1(0) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad X_2(0) = \begin{pmatrix} 1 + 10^{-8} \\ 1 \\ 1 \end{pmatrix}.$$

The recursive feedback parameters were

$$\alpha = 0.5, \quad K = 5,$$

which, by the derivation in the previous section, correspond to the effective feedback coefficient

$$\gamma_K = \frac{1 - (1 - \alpha\Delta t)^K}{\Delta t} \approx 2.4975.$$

The purpose of the following numerical experiments is to examine the effect of this recursive perturbation feedback. Each experiment compares the feedback system with the corresponding classical Lorenz simulation.

5.1 Comparison of the Attractors

The first numerical experiment compares the trajectory produced by the recursive perturbation feedback with the classical Lorenz trajectory. The objective is to determine whether the feedback noticeably changes the overall geometry of the attractor.

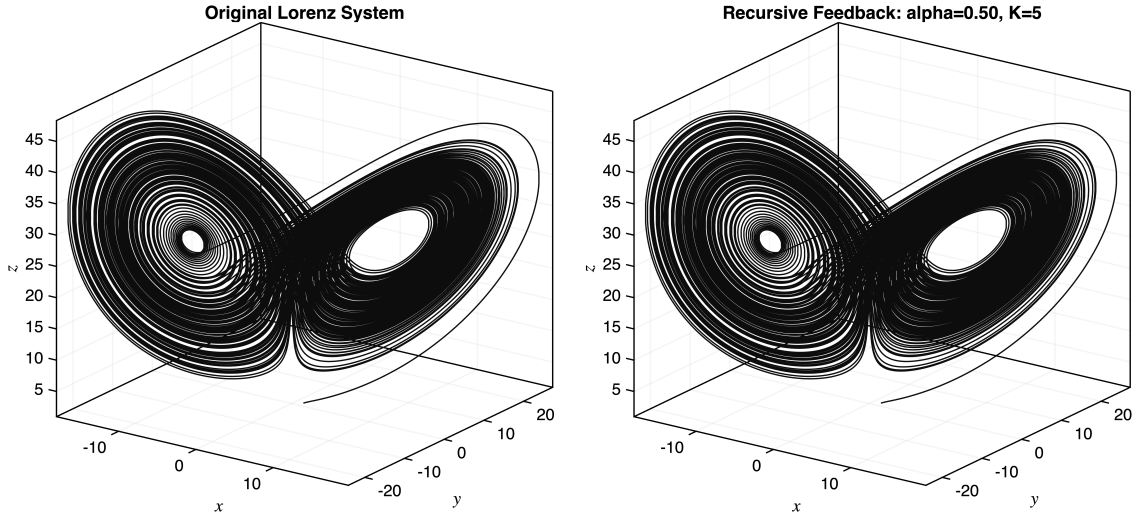


Figure 3: Comparison of the Attractors

Figure 3 shows that both trajectories exhibit the familiar butterfly-shaped structure. Although the individual paths are not identical, the large-scale organization of the attractor remains clearly recognizable after the feedback is introduced.

This suggests that recursively suppressing the separation between nearby trajectories does not, by itself, destroy the characteristic geometry of the Lorenz attractor. To understand the effect of the feedback more directly, the next experiment examines the trajectory separation itself.

5.2 Logarithmic Separation Between Nearby Trajectories

The second numerical experiment compares the separation between two nearby trajectories in the classical Lorenz system and in the recursive feedback system. Because the separation changes over several orders of magnitude, the logarithm of the distance is plotted instead of the distance itself.

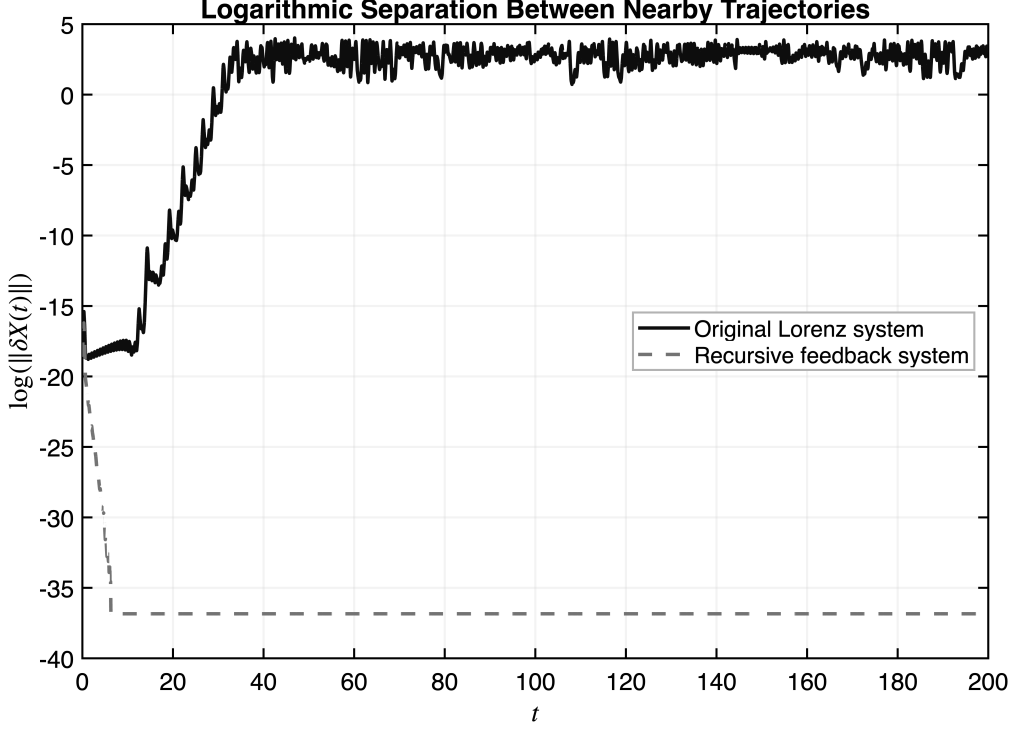


Figure 4: Logarithmic Separation Between Nearby Trajectories

Figure 4 shows a clear difference between the two cases. In the classical Lorenz system, the separation grows rapidly, reflecting sensitive dependence on initial conditions. In contrast, the recursive feedback suppresses the separation, causing it to decrease by several orders of magnitude.

This suggests that the recursive feedback is effective at reducing the growth of perturbations between nearby trajectories. The final experiment examines this behavior further by estimating the finite-time separation rate.

5.3 Finite-Time Separation Rate

The final numerical experiment examines the finite-time separation rate,

$$\lambda(t) = \frac{1}{t} \log \left(\frac{\|\delta X(t)\|}{\|\delta X(0)\|} \right),$$

which measures the average growth (or decay) of the perturbation over the finite time interval $[0, t]$.

Since the numerical simulations are performed over a finite time interval, the computed quantity should be interpreted as a finite-time estimate rather than the true Lyapunov exponent. In fact, if the limit exists,

$$\Lambda = \lim_{t \rightarrow \infty} \lambda(t),$$

then Λ is the largest Lyapunov exponent of the system.

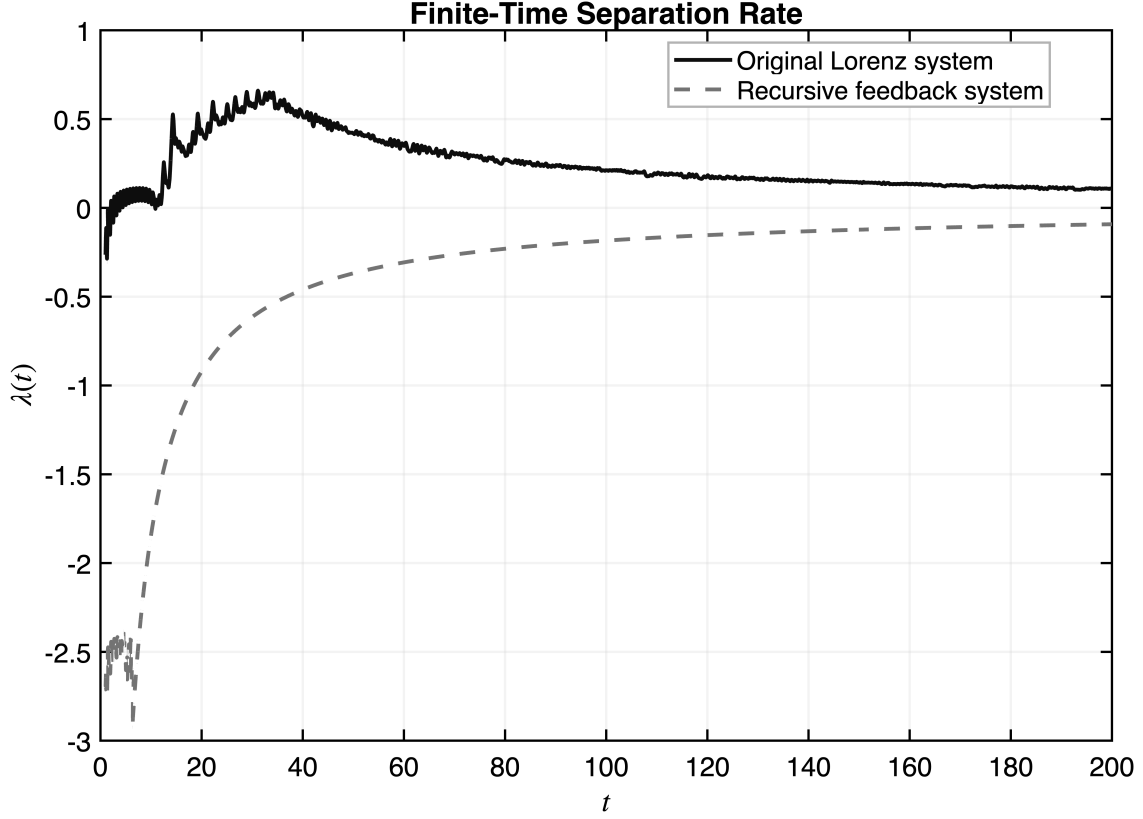


Figure 5: Finite-Time Separation

Figure 5 shows that the classical Lorenz system maintains a predominantly positive separation rate, indicating the continued growth of nearby perturbations. In contrast, the recursive feedback produces a substantially smaller, and often negative, separation rate, reflecting the suppression of the trajectory separation.

These observations are consistent with the previous experiment. Although the butterfly geometry remains largely unchanged, the recursive feedback significantly reduces the average growth of perturbations between nearby trajectories over the simulated time interval.